Unit - III Introduction to ML

3.1 History and Evaluation of ML, AI vs ML

History of Artificial Intelligence

01. **1950:** Alan Turing proposed machine intelligence through the Turing Test.
02. **1956:** John McCarthy coined the term AI at the Dartmouth Workshop.
03. **1960s–1980s:** Development of expert systems and rule-based programs.
04. **1990s:** AI started using machine learning techniques.
05. **2010s–Present:** Growth of deep learning, NLP, and generative AI.

History of Machine Learning

01. **1957:** Frank Rosenblatt developed the Perceptron model.
02. **1986:** Backpropagation improved neural network training.
03. **1997:** Deep Blue defeated world chess champion.
04. **2006:** Deep learning research gained popularity.
05. **2012 onwards:** Major advances in image and speech recognition.

Evaluation of Machine Learning

Evaluation measures how well a model performs on new data.

Classification Metrics

01. **Accuracy:** Percentage of correct predictions.
02. **Precision:** Correct positive predictions among predicted positives.
03. **Recall:** Correct positive predictions among actual positives.
04. **F1-Score:** Balance between Precision and Recall.

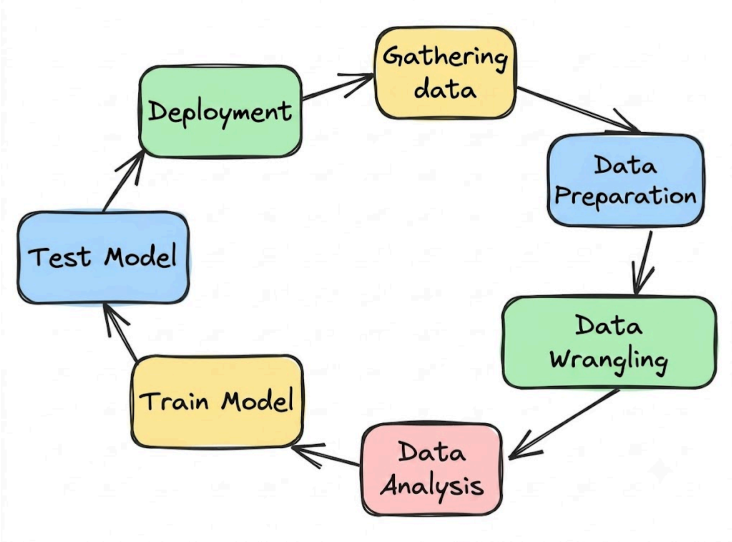
Regression Metrics

01. **MAE:** Average absolute error.
02. **MSE:** Average squared error.
03. **RMSE:** Square root of MSE.
04. **R^2 Score:** Measures model fit.

AI vs ML

AI	ML
Broad concept of machine intelligence.	Subset of AI.
Mimics human intelligence.	Learns from data.
Uses reasoning and decision-making.	Uses algorithms for learning.
Can work without training data.	Requires training data.
Example: Robotics, Expert Systems.	Example: Spam Detection, Recommendations.

3.2 Machine Learning Life Cycle



Gathering Data

Collect data from various sources such as databases, websites, sensors, and surveys.

Good-quality data improves model performance.

Example: Collect house price and location data.

Data Preparation

Clean and organize the collected data.

Remove errors, duplicates, and handle missing values.

Example: Fill or remove missing house prices.

Data Wrangling

Convert data into a format suitable for machine learning.

Perform encoding, scaling, and feature selection.

Example: Convert city names into numerical values.

Data Analysis

Study data to find patterns and relationships.

Use graphs and statistics for better understanding.

Example: Analyze how house size affects price.

Train Model

Feed training data to a machine learning algorithm.

The model learns patterns from the data.

Example: Train a model to predict house prices.

Test Model

Evaluate the model using unseen data.

Check the accuracy and performance of the model.

Example: Test predictions on new house data.

Deployment

Make the trained model available for real-world use.

Users can access predictions through apps or websites.

Example: Deploy a house price prediction system online.

3.3 Different forms of Data

Data Mining

Process of discovering useful patterns and knowledge from large datasets.

Uses techniques like classification, clustering, and prediction.

Example: Finding customer buying patterns in a supermarket.

Data Analytics

Process of examining data to draw conclusions and support decision-making.

Focuses on understanding trends and improving performance.

Example: Analyzing sales data to increase profits.

Statistical Data

Data collected for statistical analysis and interpretation.

Can be numerical or categorical.

Types: Quantitative Data: Numerical values (Age, Salary)

Qualitative Data: Categories (Gender, Color)

Example: Student marks in a class.

Statistics vs. Data Mining

Statistics	Data Mining
Studies and analyzes data using mathematical methods.	Discovers hidden patterns from large datasets.
Based on predefined hypotheses.	Finds patterns automatically.
Works on smaller datasets.	Handles very large datasets.
Focuses on explanation.	Focuses on prediction and pattern discovery.
Example: Calculate the average marks of students in a class.	**Example:** Predict which students are likely to fail based on past records.

Data Analytics vs Data Science

Data Analytics	Data Science
Analyzes existing data.	Uses data to build predictive models.
Focuses on insights and reports.	Focuses on predictions and automation.
Uses statistics and visualization.	Uses Machine Learning, AI, statistics, and programming.
Answers "What happened?"	Answers "What will happen?"
Example: Analyze last year's sales and identify the best-selling products.	**Example:** Predict next year's sales using a machine learning model.

3.4 Dataset for ML

Training Dataset

A dataset used to train the machine learning model.

Helps the model learn patterns and relationships from data.

Example: Using past house price data to teach a model how prices are determined.

Testing Dataset

A dataset used to evaluate the trained model.

Contains unseen data that was not used during training.

Example: Using new house data to check how accurately the model predicts prices.

Training vs Testing

Training Dataset	Testing Dataset
Used to train the model.	Used to test the model.
Helps the model learn patterns.	Measures model performance.
Usually forms the larger portion of the data.	Usually forms the smaller portion of the data.
Data is seen by the model during training.	Data is unseen by the model.
Example: 8,000 house records used for learning.	Example: 2,000 new house records used for testing accuracy.

3.5 Data Cleaning

Process of identifying and correcting errors in data.

Improves data quality and model accuracy.

Example: Removing duplicate records and fixing incorrect values.

Missing Data

Data values that are absent or not recorded in a dataset. Can affect the performance of machine learning models.

Handling Missing Data

a. Remove rows with missing values.
b. Replace missing values with mean, median, or mode.

Example: A student's age is missing in a record.

Name	Age
Rahul	20
Priya	Missing

Outliers

Data values that are significantly different from other values in the dataset. Can distort analysis and predictions.

Handling Outliers

- Remove outliers.
- Replace them with suitable values.
- Use statistical methods to detect them.

Example: In a class where most marks are between 40–90, a mark of 500 is an outlier.

Marks
65
72
80
500 ← Outlier